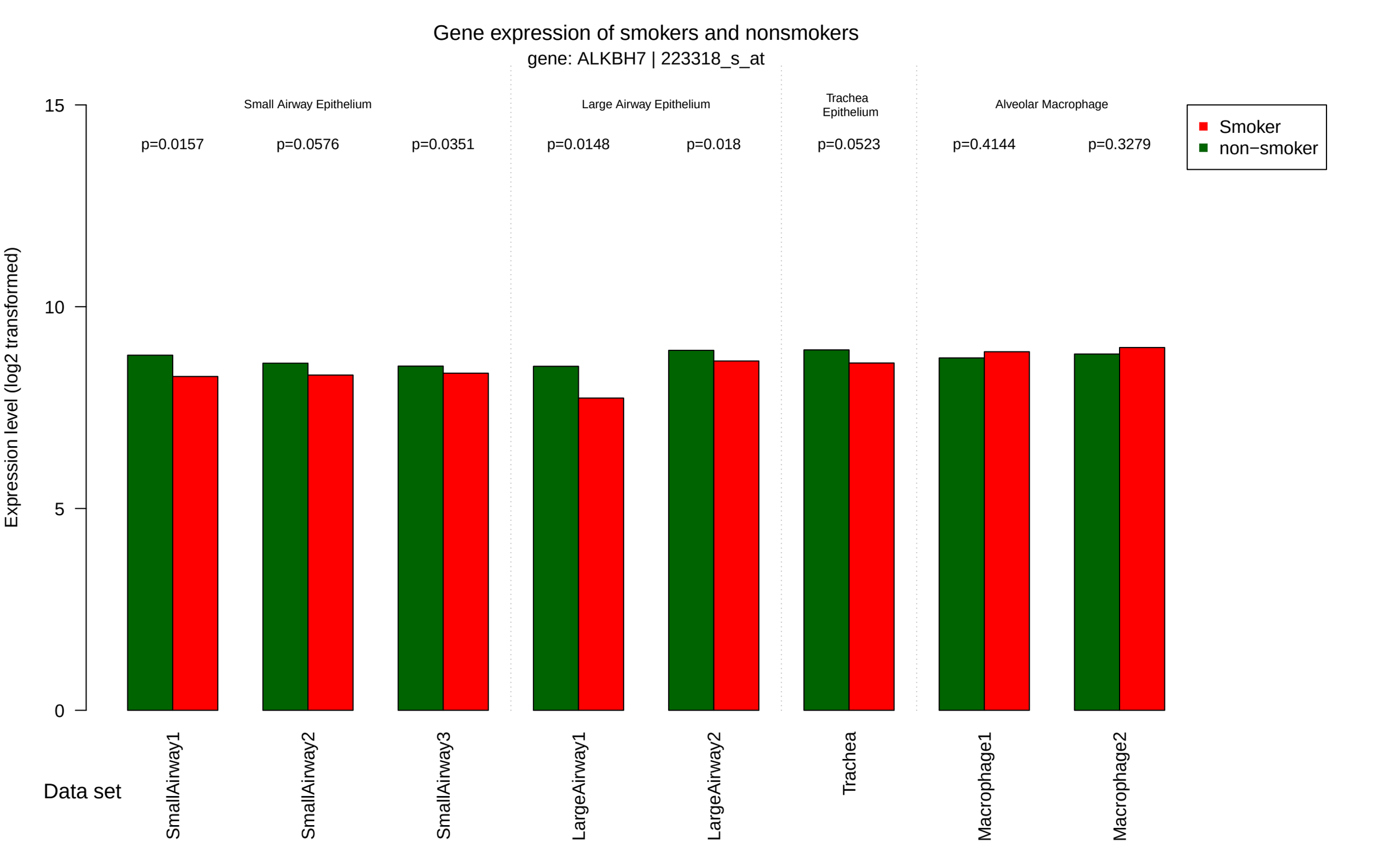
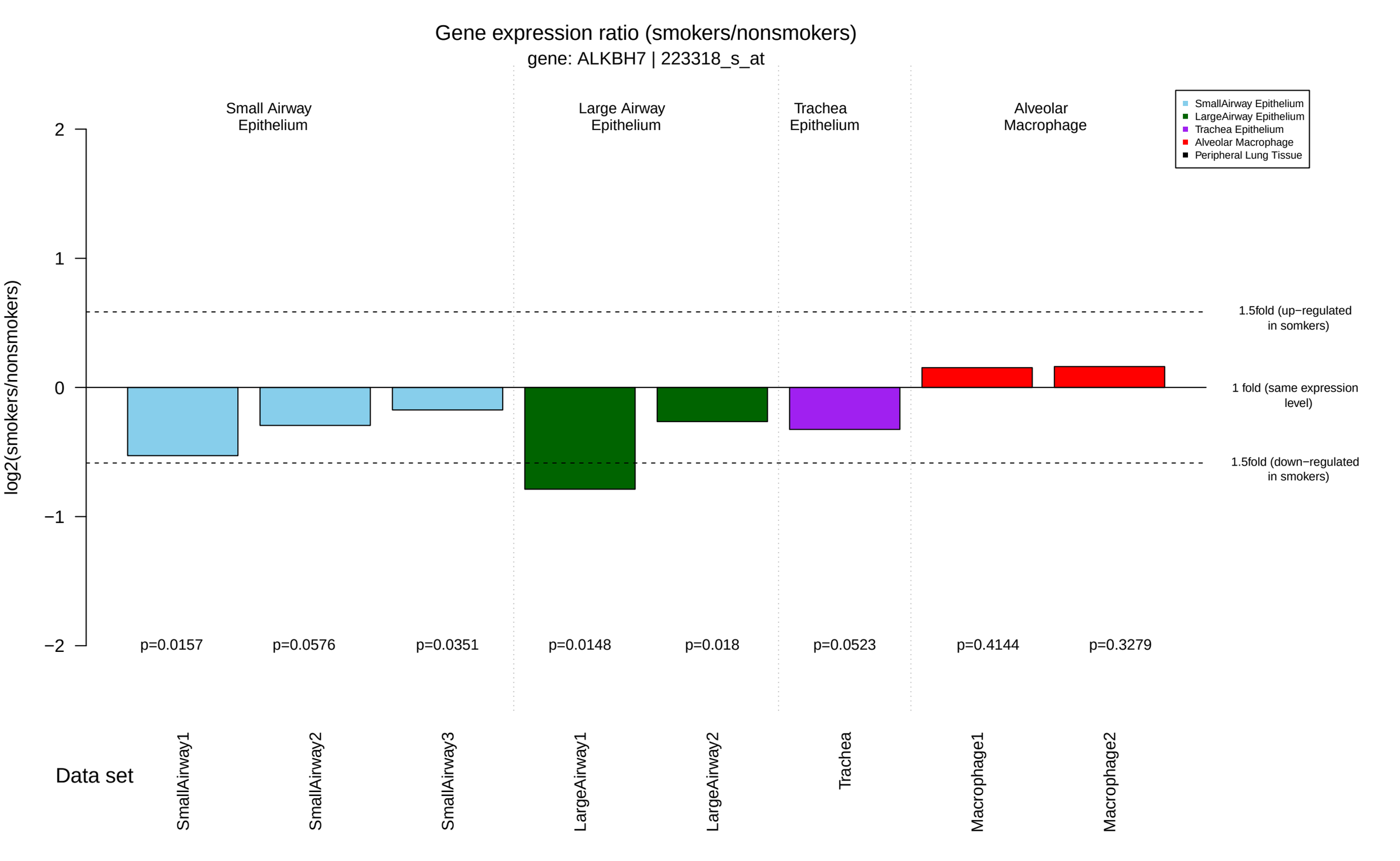
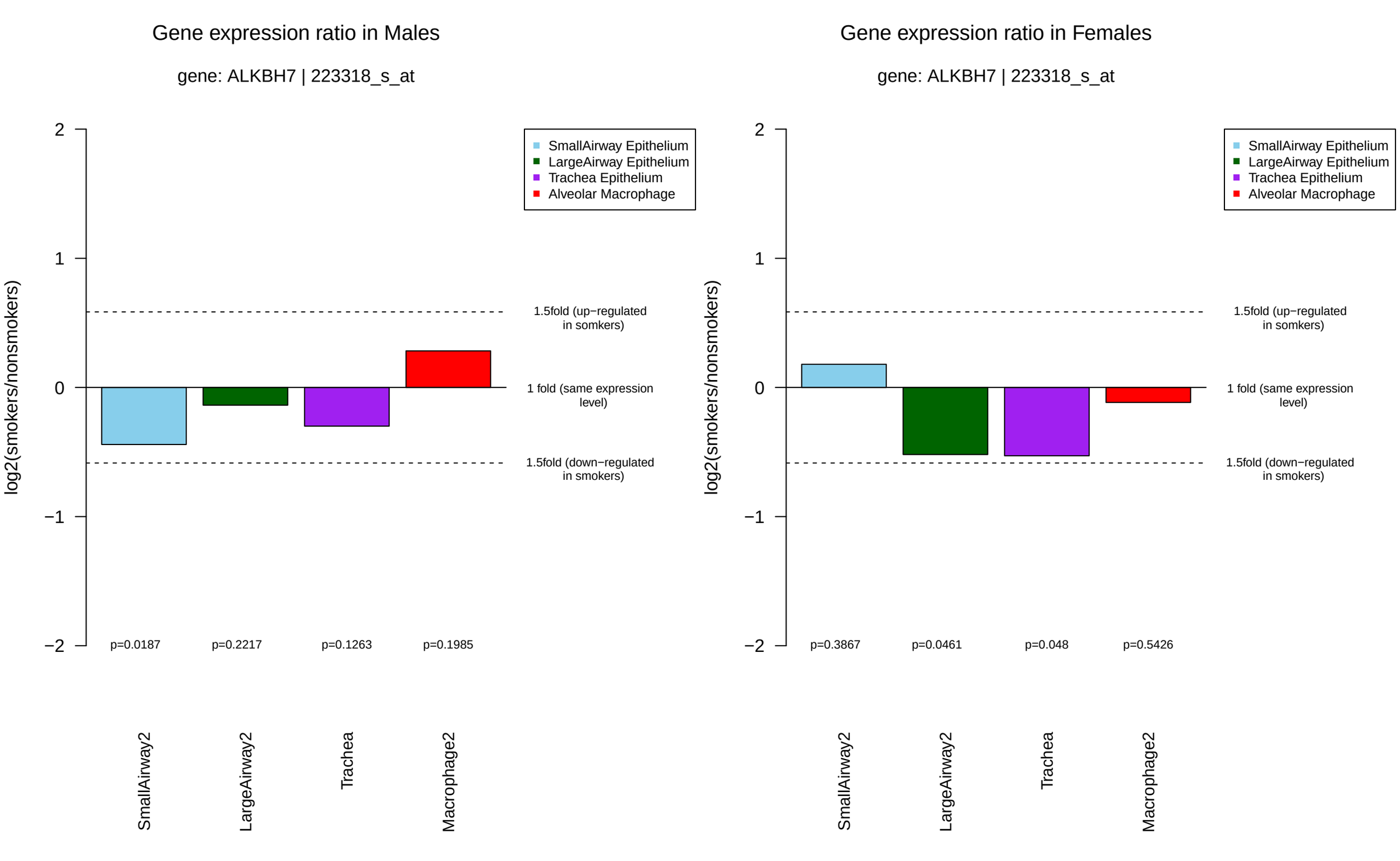
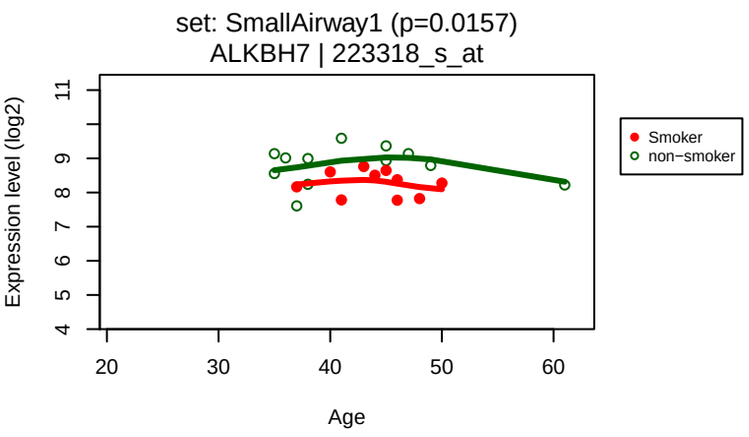
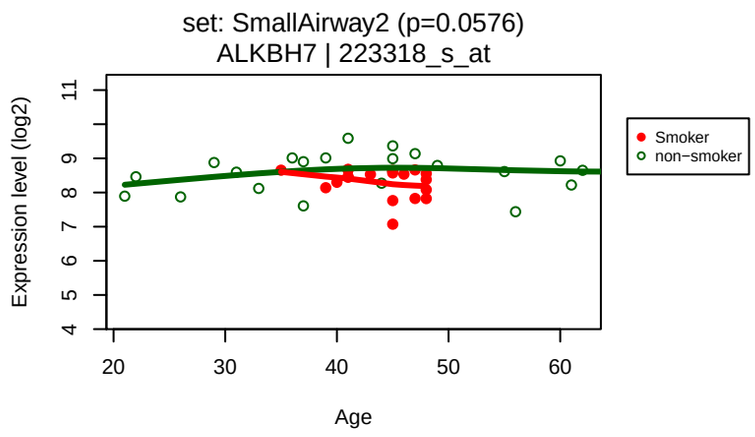
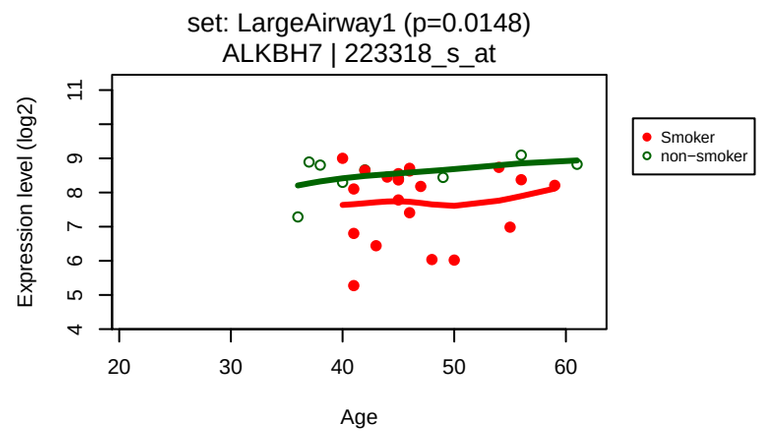

Gene expression of smokers and nonsmokers
gene: ALKBH7 | 223318_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0157
p=0.0576
p=0.0351
p=0.0148
p=0.018
p=0.0523
p=0.4144
p=0.3279
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

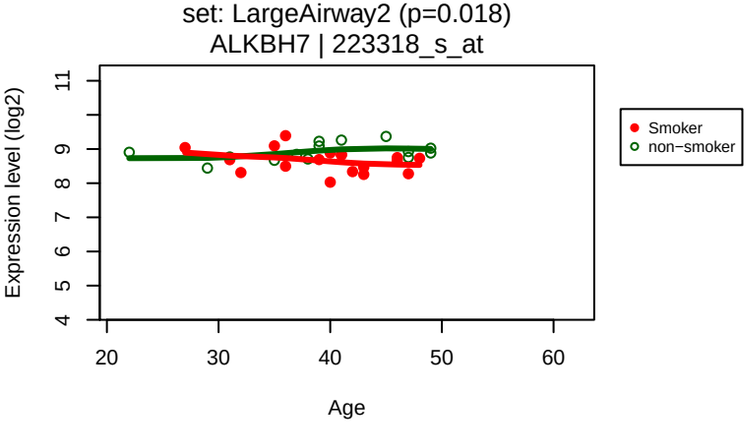

Gene expression ratio (smokers/nonsmokers)
gene: ALKBH7 | 223318_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0157
p=0.0576
p=0.0351
p=0.0148
p=0.018
p=0.0523
p=0.4144
p=0.3279
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

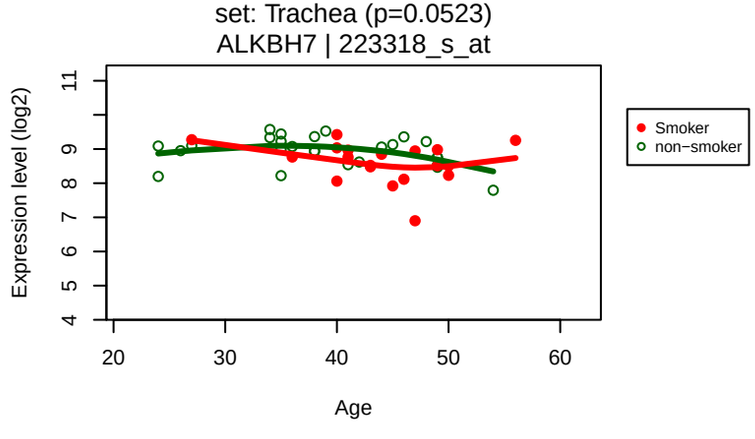

Gene expression ratio in Males
gene: ALKBH7 | 223318_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0187
p=0.2217
p=0.1263
p=0.1985
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALKBH7 | 223318_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3867
p=0.0461
p=0.048
p=0.5426
SmallAirway2
LargeAirway2
Trachea
Macrophage2

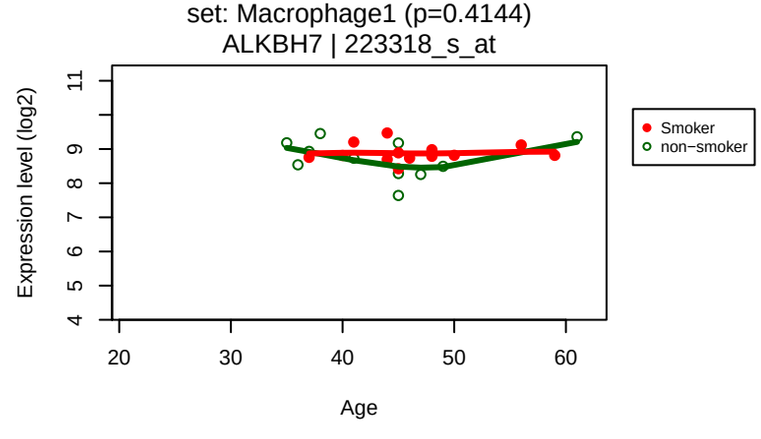

set: SmallAirway1 (p=0.0157)
ALKBH7 | 223318_s_at
Expression level (log2)
Age
Smoker
non−smoker

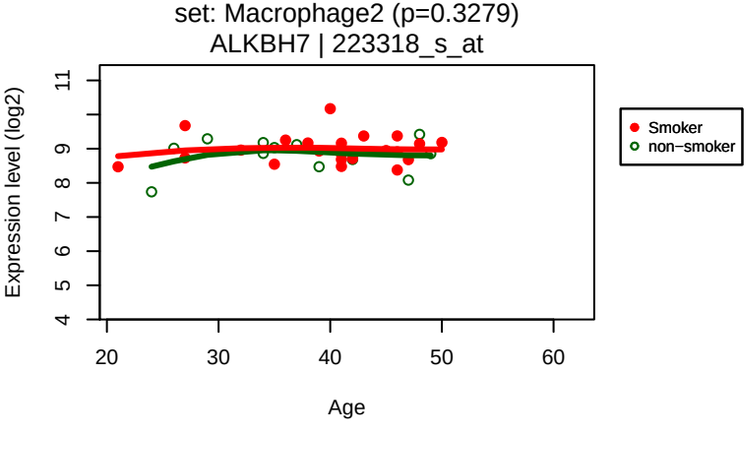

set: SmallAirway2 (p=0.0576)
ALKBH7 | 223318_s_at
Expression level (log2)
Age
Smoker
non−smoker

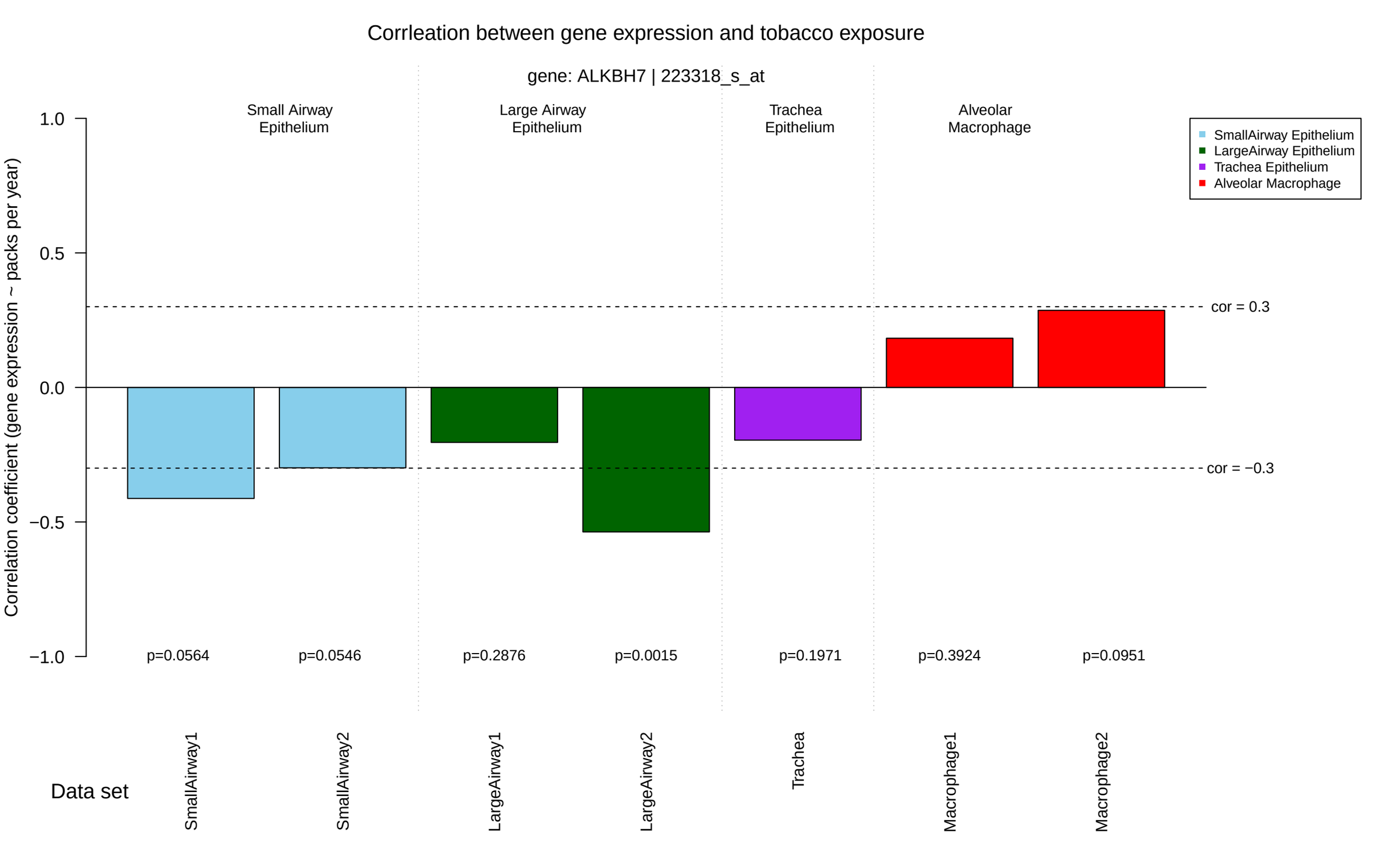

set: LargeAirway1 (p=0.0148)
ALKBH7 | 223318_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.018)
ALKBH7 | 223318_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0523)
ALKBH7 | 223318_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4144)
ALKBH7 | 223318_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3279)
ALKBH7 | 223318_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALKBH7 | 223318_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0564
p=0.0546
p=0.2876
p=0.0015
p=0.1971
p=0.3924
p=0.0951
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2